



SOCIOECONOMIC DEVELOPMENT & HOUSEHOLD WELL-BEING IN SOUTH AFRICA (SDHSA)



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Project case study

In South Africa, many households lack full access to basic services. This lack of access is closely tied to poverty. Financially challenged households are usually the ones missing these services. Government departments have limited money to spend on infrastructure, so they need to know which service gaps matter the most, so that they can spend that money where it will be of help to most people.

Decision problem

Determine which gaps in access to water, electricity, and sanitation are most strongly associated with household poverty in South Africa. This will help government departments, NGOs, and policymakers prioritize the most urgent services and allocate infrastructure budget resources where they can have the greatest impact.

Investigative Questions

- Which basic services: electricity, water, and sanitation are most strongly associated with household poverty?
- which should be prioritized for intervention?

Objectives

- Identify the relationship between basic services and poverty.
- Produce evidence that supports Infrastructure investment decisions.

Process

We combined five datasets covering 20,925 households, cleaned and checked them, and compared households. We then built three different prediction models to see which household feature best associates the two groups and tested them on untrained household data.

Findings

We found that no single basic service has the strongest association to poverty, however of the three services, Sanitation is the most consistent signal. Electricity shows a large gap, but it affects few households. Water safety shows no link to poverty in this dataset. The selected model identifies about 56 of every 100 low-income households (sensitivity 55.6%). About 26% of surveyed households are in the Low-income group.

If the department wants to start with one service, sanitation is the best-supported starting point, but the evidence does not show that fixing a service would, by itself, lift a household out of poverty. These are patterns in a one-off survey, not proof of cause and effect.

Technical Document

This Technical document will be structured according to the CRISP-DM process undertaken to find the results/insights.

Business understanding

Many South African households lack full access to electricity, clean water and proper sanitation, and these gaps are closely tied to poverty. Because departments cannot fix everything at once, they need to know which gap matters most

Business Question

How can NGOs and policymakers prioritize investments in water, electricity, and sanitation services to reduce household poverty most effectively in South Africa?

Data understanding

Data Sources

These 5 datasets were provided by our Project Case Study material.

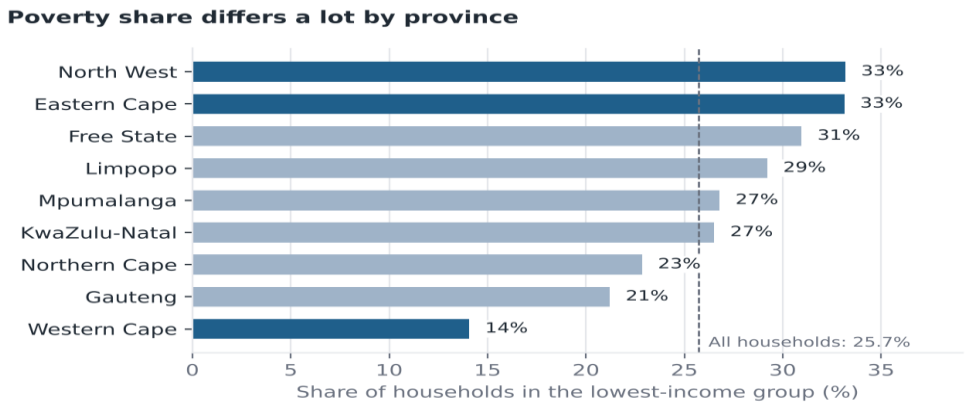
Dataset	Content
D05 Basic Services	Water, electricity and sanitation access
D06 Household Energy	Electricity access and load-shedding frequency
D07 Household Wellbeing	Income, hunger, asset ownership
D09 Household Resources	Vehicles, household size, income source
D04 Dwelling Conditions	Wall/roof/floor condition, tenure, RDP housing, subsidy

`Data exploration and joining`

HouseholdID was used as the primary key across all five datasets. Uniqueness and row counts were verified before and after each inner join to track and record households removed; the resulting analytical dataset contained 20,925 households, of which 20,475 had usable income data.

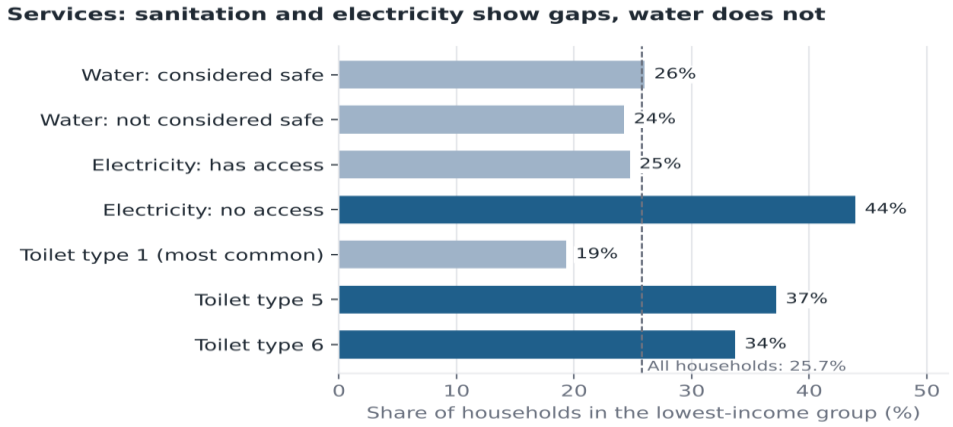
Income is highly unequal. The median household income is R5,030 per month, compared with a mean income of R11,752, which is influenced by a small number of high-income households, including the highest recorded income of R1.46 million. 450 households without usable income were excluded from the poverty analysis and not estimated.

Poverty also varies by location. The North West and Eastern Cape have the highest proportion of households in the poorest group, at approximately 33%, while the Western Cape has the lowest, at about 14%.



Population figures are weighted by survey household weights (24.9% Low income). Sample figures and model metrics are unweighted (about 26%).

Services and poverty:



- **Sanitation:** shows a clear relationship with poverty.
- **Electricity:** reveals a substantial poverty gap. However, this issue affects a relatively small proportion of the sample.
- **Water:** shows no meaningful relationship with poverty. Households whose water was considered unsafe had a slightly lower poverty rate than those with safe water access. This unexpected pattern suggests that water access, as measured in the data, is not a strong indicator of household poverty.

Data preparation (EDA)

Exploratory analysis

During the EDA process, we used a problem -> solution -> result framework.

Problem	Treatment	result
Records whose household ID read “UNMATCHED” instead of a number	Removed those 15 records but recorded	Every remaining ID is a valid household number

Problem	Treatment	result
Duplicate households	Kept the first copy of each; the original count of records was recorded	Exactly one identifier key per household
Income recorded as a "placeholder" such as 9,999,999, or larger (the largest raw value was 10 billion)	Marked as "missing" rather than guessing or treating them as zero	Income now holds only real values
Impossible toilet code, household size 0 and text vehicle counts	Marked as "missing" (the household is kept, only the bad value is blanked)	Households are not lost because of one invalid entry
Province spelled many ways	Mapped every spelling to the nine official names	Province comparisons are reliable
Missing values	450 households without usable income were excluded from the poverty analysis and not estimated. Missing values in other predictors were median-imputed, with values fitted on the training data only.	Less bias results in our prediction models distribution results.

Feature engineering

Redundant features that provided little or no additional value to the model were removed. After integrating the datasets, the feature set was reduced to 17 variables to maintain model interpretability and present clear, relevant insights in response to the business questions.

predictors	**Feature/variable name**	**Description**
Target	Target	Provides the two classes required for binary classification
Water predictor	WaterSafe_label	represents household water access
Electricity predictor	ElectricityAccess_label	represents household electricity access
Sanitation predictor	MainToiletCode_clean	Represents the household's main toilet/sanitation type
Composite score	ServiceAccessScore excluded	Combines the three individual services and would hide which service is most important

The final feature set was designed to remain interpretable, avoid redundancy, and align directly with the business objective. Income, log-income, and other income-derived variables were excluded to prevent the model from predicting poverty using the same measure that defines it. Household IDs and survey weights were also removed.

Modelling

After the exploratory data analysis, the dataset was divided into training and testing sets using an 80/20 split. The training set contained 16,379 households, while 4,096 households were reserved as an unseen test set to evaluate the model's performance.

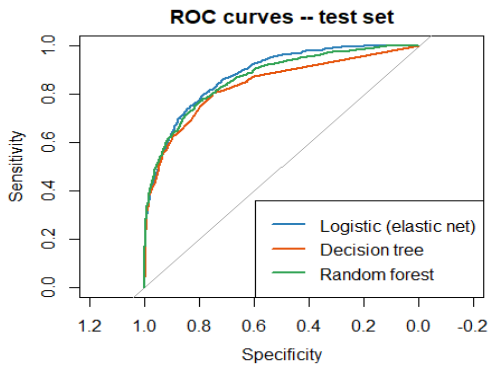
We kept reproducible results best practices [set seed (381)]

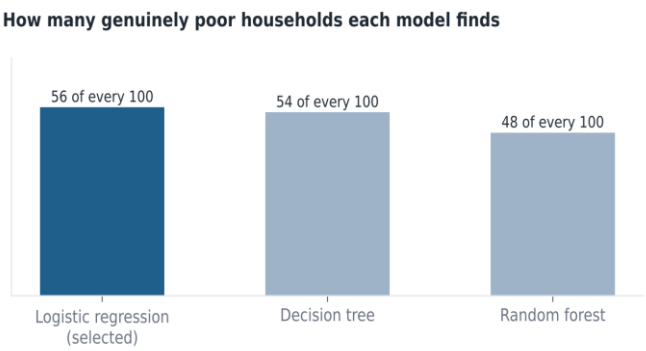
Model Selection

Regression on raw income was unsuitable because it predicts exact income values rather than whether a household is poor. Clustering was also inappropriate because it identifies groups without a predefined outcome. Therefore, **binary classification** was chosen to distinguish between **low- and non-low-income households**. Three candidate modelling techniques were used, with service-related predictors evaluated using sensitivity and ROC-AUC to determine which basic service is most strongly associated with low-income households.

Modeling technique	**Reason**
Logistic regression (elastic net)	Can use its coefficients to show the strength and direction of each service's association with the Low class. Logistic regression finds the poorest households and separates the groups best. This is the recommended tool, and the reason we can say anything about individual services.
Decision tree (CART)	can use variable importance to show which service contributes most to classification
Random Forest	The random forest is marginally the most "accurate", but it misses more than half of the poor households. Missing a household that genuinely needs help is a costlier mistake than flagging one that turns out not to.

Results





Tool	Overall correct	Poor households found	Others correctly cleared	Separating ability (AUC)
Logistic regression (selected)	83.4%	**55.6%**	93.0%	**0.879**
Decision tree	83.3%	54.1%	93.5%	0.839
Random forest	83.9%	48.1%	96.3%	0.866

The analysis finds that sanitation is the most consistent indicator among basic services, but household income and assets are stronger measures of poverty. Regression and k-means clustering were used as robustness checks to test whether service and wellbeing profiles align with income.

Evaluation

Because only about 26% of households belong to the Low class, accuracy alone could give a misleading impression. A model that classified every household as Not Low would still achieve about 74% accuracy but would fail to identify households needing attention. The analysis therefore also reported Low-class recall, specificity, balanced accuracy, ROC-AUC, and confusion matrices to provide a more complete assessment of performance.

Model	Accuracy	Sensitivity (Low)	Specificity	Balanced Accuracy	ROC-AUC
Logistic (elastic net)	0.834	**0.556**	0.930	0.743	**0.879**
Decision tree	0.833	0.541	0.935	0.734	0.839
Random forest	**0.839**	0.481	**0.963**	0.722	0.866

Elastic net logistic regression was selected as the preferred model. Although random forest achieved slightly higher accuracy and specificity, its sensitivity was only 48.1%, meaning it failed to identify more than half of households in the Low class. Because overlooking an under-served household is more costly than generating a false alert, elastic net was preferred for its stronger sensitivity, ROC-AUC, and balanced accuracy.

Table X: Recall for Low-income households by province (selected model, test set)

Province	Test households	Actually Low	Recall
Eastern Cape	505	172	0.657
North West	260	94	0.628
Free State	246	75	0.613

Mpumalanga	347	90	0.578
Limpopo	485	152	0.566
KwaZulu-Natal	628	163	0.552
Gauteng	1090	214	0.472
Northern Cape	173	43	0.442
Western Cape	362	51	0.392

Recall ranges from 0.657 in Eastern Cape to 0.392 in Western Cape. Western Cape, Northern Cape and Gauteng are the weakest. Gauteng matters most because it also holds the largest number of low-income households. Only province was checked.

Findings

Population figures use survey weights, as on the dashboard. Model metrics use the unweighted test set (Table X).

Model performance. Elastic-net logistic regression was selected because it had the best sensitivity (0.556) and ROC-AUC (0.879) of the three models. Selection reasoning is in the Evaluation section.

Services. No single service is the dominant signal. Of the three:

- **Sanitation** is the most consistent. The Low-income share is about 39% for toilet type 5 and about 35% for types 6 and 7, against 18% for the most common type.
- **Electricity** shows a large gap (about 44% Low income without access, 24% with access), but only about 5% of households lack access, and the model found it weak once other factors were included.
- **Water safety** shows no link: 23.5% where water is “not considered safe” and 25.1% where it is “considered safe”.

Household characteristics. Asset ownership and household size are stronger signals than any single service, and the Low-income share falls steadily as the asset index rises. Income source also predicts strongly, but it overlaps closely with income, so we do not treat it as an independent explanation. These are predictive associations in survey data. They do not show that a service condition causes poverty.

Provinces. About 24.9% of households are Low income. The rate is highest in North West and Eastern Cape (about 33%) and lowest in Western Cape (14.3%). The largest numbers of low-income households are in Gauteng (about 1.15 million), KwaZulu-Natal (about 0.90 million) and Eastern Cape (about 0.58 million), together about 2.6 million of

4.7 million. Ranking by rate and ranking by number of households give different priorities.

Evidence-based Recommendations

Evidence	Recommendation	Caution
Sanitation is the most consistent service signal	Prioritise sanitation, starting with the toilet types with the highest Low-income shares	Association only. It does not show that upgrades lift households out of poverty
Highest rate: North West and Eastern Cape (about 33%). Most households: Gauteng and KwaZulu-Natal	Decide whether targeting is by rate or by reach, then set provincial priorities to match	The two measures rank provinces differently
Electricity: about 44% vs 24% Low income, about 5% of households	Treat as a smaller, targeted second priority	Weak after other factors are included
Water shows no link, but "safe" is self-reported	Re-test water with better measures (interruptions, distance) before deciding	A perception, not a water-quality test
Sensitivity 0.556. Recall 0.392 to 0.472 in Western Cape, Northern Cape and Gauteng	Use the model as a screening aid. Add human review and local data in those provinces. Do not treat a "NotLow" prediction as grounds to refuse support	Misses about 44% of low-income households.
Unlabelled toilet and income codes	Obtain the codebook and cleaner ID and income codes before the next survey wave	Improves reliability

Service gaps should be used together with asset ownership and household size, which are stronger signals than any single service. The findings guide prioritisation and should not decide eligibility on their own.

Limitations and uncertainty

- **False negatives.** The model misses about 44% of low-income households in the test set and cannot be assumed to find everyone who needs help.

- **Uneven provincial performance.** Recall is lowest in Western Cape (0.392), Northern Cape (0.442) and Gauteng (0.472). Only province was checked, not urban/rural or household size.
- **Association, not causation.** One-time, self-reported survey data cannot show whether service access changes household poverty.
- **Definition of "Low".** It means the bottom 25% of surveyed households, not an official poverty line, and income is per household, not per person.
- **Missing data.** 450 households without usable income were excluded, and other predictors were median-imputed. These choices can influence estimates.
- **Overlapping variables.** Income source overlaps with income, and assets partly reflect income, so neither is an independent explanation.
- **Incomplete codebook.** Some codes (for example toilet types and income source) cannot be interpreted confidently. Very small categories, such as toilet types 10 and 12 (22 and 57 households), are unreliable.
- **Weighting.** Dashboard figures are weighted and model metrics are not, so they are not directly comparable.
- **Generalisation.** Test-set performance does not guarantee performance on a later survey wave or another population.

Future work

Collect per-person income, complete the codebook, track households over time, test better water and sanitation measures, test a lower decision threshold to catrch more low-income households (at the cost of more false alarms), check fairness by urban/rual and household size, review the model quarterly, with retraining when the thresholds in the *Monitoring strategy* are met.

Deployment

Deployment strategy

Intended users - Project/NGO staff and policy analysts working on the SDHSA initiative who need to prioritise which households or areas to review for service-access interventions, not data scientists.

Decision supported – Whether a household (or a batch of households from a new data collection) should be flagged for further review as likely to be in the Low-income group, so limited field-visit or intervention capacity can be prioritised.

Inputs – The same 17 predictors used in Task 1 (province, water/electricity/sanitation access, household size, asset index, dwelling conditions, income source, etc.), entered

either one household at a time through a form, or as a CSV of many households at once.

Outputs – A predicted class ("Low" / "NotLow") and the model's estimated probability of "Low", together with an on-screen caution when a prediction falls in a province (Western Cape, Northern Cape) where Task 3's fairness check showed lower reliability.

Operational workflow – New or updated household records → uploaded as CSV or entered individually → validated against the required column set → missing/unseen values handled the same way as in training (median imputation for numeric fields, an explicit "Missing" category for categorical fields) → scored by the selected elastic-net model → results reviewed by a human analyst before any resource-allocation decision is made. The model is a screening aid; it does not make the final decision.

Tool	Fit for this project
R/Shiny	Runs the trained "caret"/"glmnet" model natively with no re-implementation; built-in web form for non-technical users; supports both single-record and batch-CSV workflows in one app. **Selected.**
Plumber (R API)	Good for system-to-system integration, but would need a separate front end for the non-technical users this project targets - more effort for no real benefit here.
flexdashboard	Suited to static/periodic reporting dashboards rather than an interactive "score this household" tool with user input.
Power BI + R script	Possible, but ties deployment to a Power BI licence and is a worse fit for ad-hoc single-household scoring than a purpose-built form.

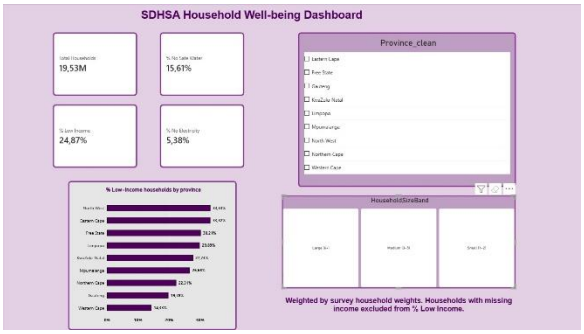
Monitoring strategy

Measure	Threshold/indicator	Action
Sensitivity (recall for "Low")	Falls below ~0.50 on new labelled data (current test-set value: 0.556)	Flag for retraining

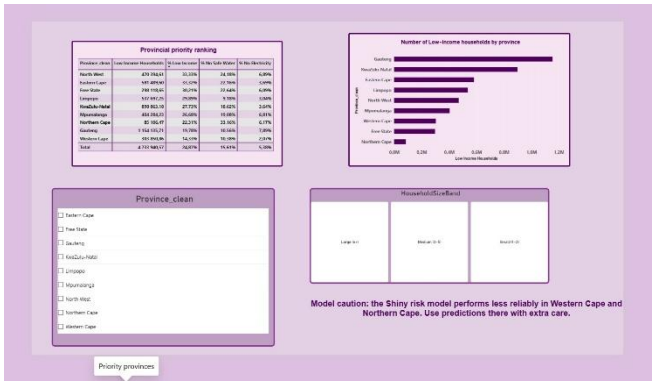
The model should be reviewed at least quarterly using newly labelled data. Retraining should be considered if ROC-AUC falls below 0.80, while noticeable changes in province, asset, or dwelling-condition patterns should trigger further investigation. A new survey wave requires retraining and re-validation, and changes to variable definitions require the current model to be retired and redeveloped.

Power BI Decision-Support Dashboard

The dashboard allows planners to explore where low-income households are and which gaps go with them, using the final analytical dataset (20925 households). All percentages are weighed by survey household weights, and the 450 households with missing income are excluded.







Key insights: 24.9% of households are low-income. North West and Eastern Cape have the highest share (about 33%), but Gauteng (about 1.15 million) and KwaZulu-Natal (about 0.90 million) have the most low-income households, so targeting by share and by scale give different priorities. The low-income households share falls as service access and asset ownership rise, and households without electricity are low-income about 44% of the time against 24% with access. Water safety shows no meaningful link. The dashboard shows associations, not causes.